

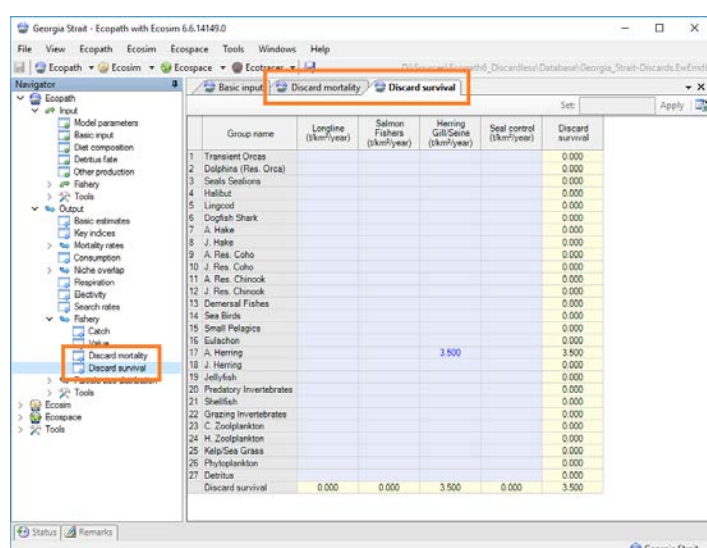
Discards accounting in EwE

Implemented for Discardless / MINOUW by Ecopath International Initiative

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Ecopath

Internally, Ecopath was already accounting for discards, but this accounting was not exposed to the users. New Ecopath output tables have been added that provide landings, discard mortality and discard survival estimates, by fleet and group (Figure 1). These can be accessed via Navigator > Ecopath > Output > Fishery.



Note that the Azores team has expressed the need to add a column to the discard fate table to allow part of the unwanted catch to be landed, and that possibility should be given to set an economic value to this part of the landings different from the value of the wanted landings. It is understood that valuing exported discard fate (representing the value of unwanted catch) is necessary to explicitly model the implications of EU landings obligations. The current EwE setup is able to account for this costing in an indirect way by splitting fleets, etc.

Figure 1 - New Ecopath discards tables

Implementation uncertainty is high regarding this point, mostly because the Ecosim and Ecospace effort gravity models will be greatly affected. A proper implementation is estimated to take 1 to 2 weeks to implement, and a week to test. The Discardless budget is insufficient to address this change. EwE researchers at Stockholm University have expressed interest in writing a follow-up proposal to implement this specific feature.

Ecosim

A series of discards accounting changes have been made to Ecosim.

Exposing existing calculations in the UI and CSV output

The internal accounting for discards and landings within Ecosim is now exposed to the EwE software user as new discards-related graphs (Discards mortality and Discards survival) to the Ecosim group plots (Figure 2), and in the Ecosim CSV output. The new group plots can be found under Navigator > Ecosim > Output > Ecosim group plots. In addition, a fleet-centric series of plots have been added to Ecosim to allow exploration of Ecosim results by fleet (Figure 3). These plots can be found under Navigator > Ecosim > Output > Ecosim fleet plots.

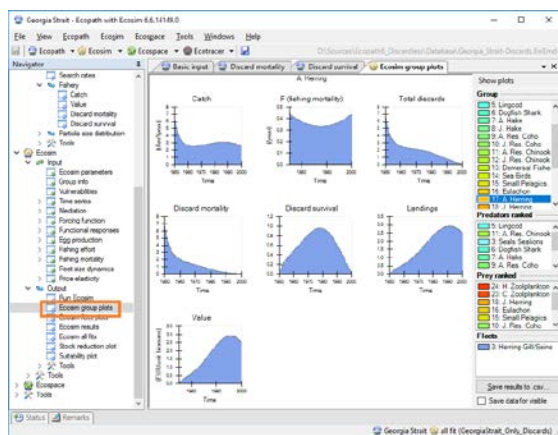


Figure 2 - New Ecosim discard plots, per group

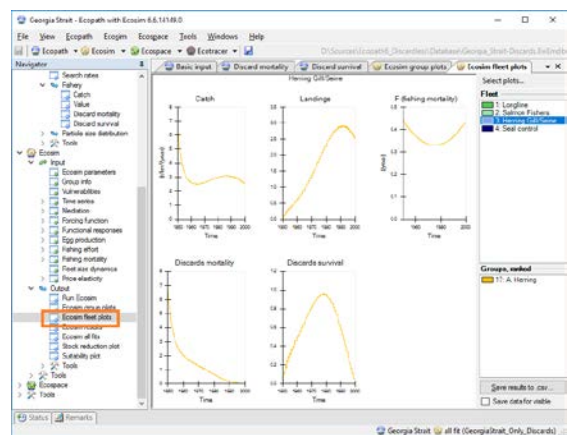


Figure 3 - New Ecosim discard plots, per fleet

Allow varying discards in Ecosim

The ability to represent variations in discard strategies, either by varying the proportion discarded, or by changing discard mortality, were added through two new types of absolute driver time series: Discard Proportion (type 10) and Discard Mortality (type 11) (see Figure 4).

Discard Proportion time series values provide a multiplier (for a given fleet and group) over time, to the base discard proportion as entered in the Ecopath input parameters.

Similarly, Discard Mortality time series values provide a multiplier (for a given fleet and group) over time, to the base discard mortality rate as entered in the Ecopath input parameters.

These types of time series introduce two new features to already existing time series types:

1. Discards are specified by fleet and group. As such, the time series requires specification of a fleet (first pool code) and a group (second pool code).
2. To be able to entirely shut down discarding, both new time series types accept 0 as an actual value to entirely shut down discarding. To make sure Ecosim does not force discards for a given year, simply provide an empty cell in the time series CSV file (see Figure 4, last column. last visible rows)

The Ecopath user interface has been updated throughout to consistently reflect the double pool codes for the new types of time series.

Figure 4 – Importing the new discards driver time series into Ecosim

Using landings and discards for model fitting

Ecosim has gained the ability to include reference time series of landings and discards.

The Landings time series (type 13) is a reference time series that specifies the absolute landings of catches of a given functional group by a given fleet, over time. Landings time series accept 0 as valid values.

The Discards time series (type 14) is a reference time series specifies the total discards (regardless of discard survival) of a given functional group by a given fleet over time. Total discards were used for this type of time series because statistics for total discards are easier obtained than discard survival or mortality data. Discards time series accept 0 as valid values.

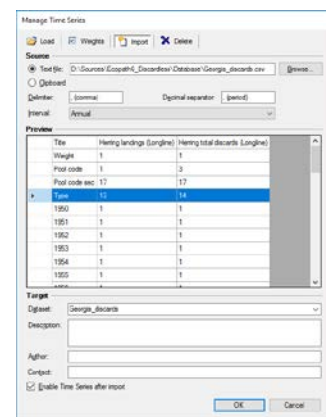


Figure 5 - Importing the new landings and discards reference time series

New fitting-related graphs (Landings and Total discards) were added to the Ecosim group plots (Figure 2). Both time series types have been included in the Sum of Squares calculations for the model fit, and the Ecosim fit to time series plots will display these time series when present.

Ecospace

The spatial-temporal model has been extended to optionally incorporate Ecosim discards forcing into its calculations. To enable Ecosim discards forcing in Ecospace, go to Navigator > Ecospace > Input > Ecospace parameters, and under the Time Series heading at the bottom, check “Use Ecosim discards forcing” (see Figure 6).

Please note that this option will only be enabled if Discards driver time series are present, and that Ecosim discards forcing in Ecospace is NOT enabled by default.

The Ecospace run form, where Ecospace is executed, has been extended with the display of total discards maps per functional group. These maps can be written to disk during an Ecospace model run as part of the Ecospace auto-save logic (Figure 7).

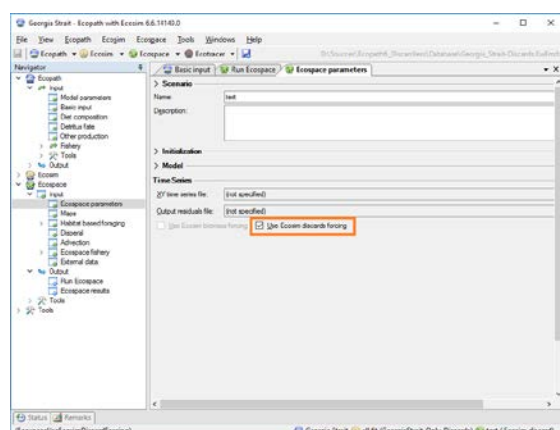


Figure 6 - Enabling Ecosim discard forcing in Ecospace

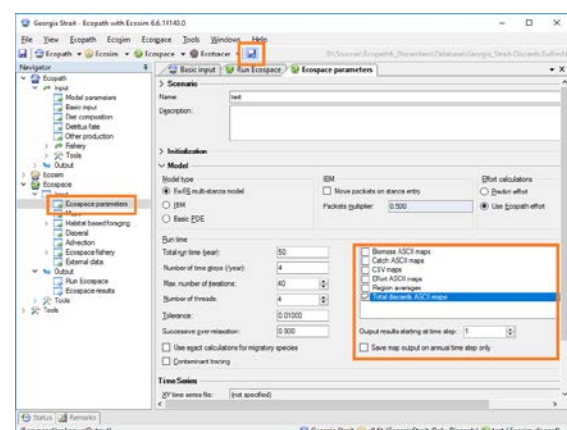


Figure 7 - Enabling automatic Ecospace discard map saving

Management Strategy Evaluations

The new Ecosim discards driver logic may conflict with the internal workings of the two Management Strategy Evaluation (MSE) routines in EwE, which already contain sophisticated discard management and assessment logic. In order to avoid conflicts with these routines, the user is now prompted of this potential conflict when running either MSE routine (Figure 8).

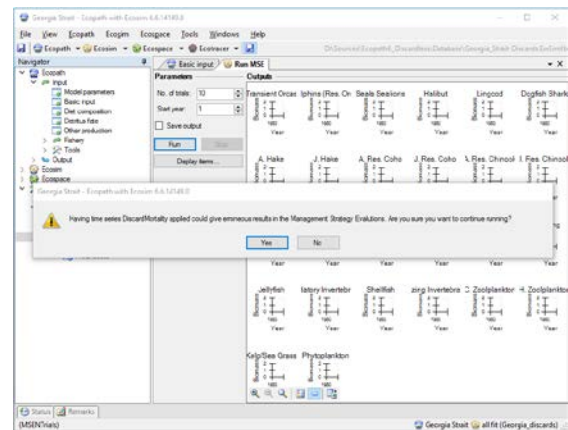


Figure 8 - New MSE discards prompt